How To Minimise Electromagnetic Interference And Improve Compatibility In MEDICAL DEVICES

This article talks about methods to improve EMI and EMC in medical devices

Multiple medical electronic devices working simultaneously in a surgery room

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Electromagnetic compatibility (EMC) of a device means that it is compatible with its electromagnetic environment and does not emit electromagnetic waves to a level that causes electromagnetic interference (EMI) in other devices. An increase in emission of electromagnetic energy of a device beyond a limit can disturb its own operation or of an electronic device operating nearby. So, manufacturers must design their devices with adequate protection against possible interferences.

The difference between immunity and emission testing is that while immunity requirement is concerned with how susceptible a device is to electromagnetic energy emitted from surrounding devices, emission requirement is concerned with the amount of electromagnetic energy emitted from the device.

IEC 60601-1-2 is the EMI/EMC standard for medical electrical equipment and systems. Medical devices require complex analysis of EMI/EMC regulations since these devices are no longer used in just a hospital setting. With the increasing use of wearable medical technology, patients can be anywhere in the world in any environmental condition. EMC standards must comply for improved designs of medical devices.

Importance of EMI shielding

Due to the increase in demand for medical electronic devices, development in this sector is happening fast. Resolution level is continuously increasing to provide greater details and more accurate diagnosis. To ensure an enhanced level of detail, it is necessary to create an environment where EMC and EMI do not interfere with the operation of medical electronic systems. Imaging devices must provide reliable and noise-free data even in harsh environments, like around MRI or X-ray systems.

EMI can fail monitoring systems or life-support equipment. Patients and